gas may be zero in this cycle, the internal energy is not zero.

12 Fig. 12.1 shows the variation with displacement *x* of velocity *v* of a simple harmonic oscillator. Fig. 12.2 shows the variation with time *t* of the acceleration *a* of the oscillator.

Which of the points on the *a*-*t* graph corresponds to the state of motion represented by point **P**?

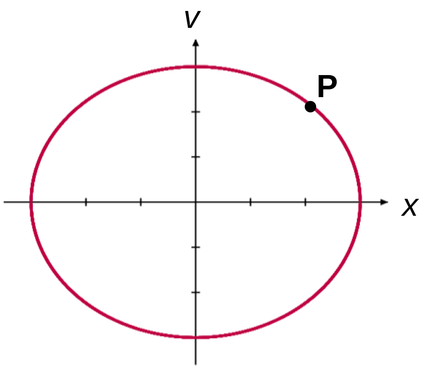

Fig. 12.1

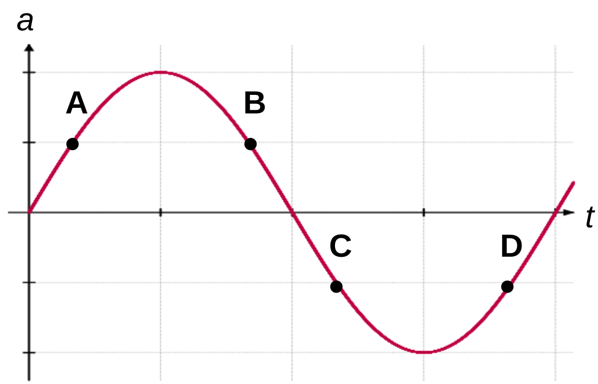

Fig. 12.2

C At point P, the displacement and velocity is positive.

Using $a = -\omega^2 x$. The *a* should be Negative (A and B no longer plausible)

For *v*, sketch *x*-*t* graph and find the point of positive velocity (gradient of *x*-*t* graph).

13 An object that undergoes simple harmonic motion has an amplitude A and total